

Core Coverage and Exact Computation for Three-Dimensional Queen Domination

Mahesh Ramani

17 August 2026

Abstract

Let Q_n^3 be the queen graph on the $n \times n \times n$ board. For the inner core $C = \{1, \dots, n-2\}^3$, we determine the number of vertices of C dominated by a queen at a corner, on an edge, on a face, or in the interior. In particular, the best boundary placement has strictly smaller core coverage than a central interior placement. The calculation is compatible with a reduction under the order-48 octahedral symmetry group.

The elementary projection, volume, and layer bounds give

$$\max \left\{ \gamma(Q_n^2), \left\lceil \frac{n^3}{13n-12} \right\rceil \right\} \leq \gamma(Q_n^3) \leq n\gamma(Q_n^2) \leq \frac{69}{133}n^2 + O(n),$$

and hence $\gamma(Q_n^3) = \Theta(n^2)$. An integer linear programming computation, with symmetry reduction and independent solution checks, gives

$$\gamma(Q_n^3) = 1, 1, 1, 4, 6, 8 \quad (n = 1, \dots, 6).$$

For $n = 7$, the certified bounds are $10 \leq \gamma(Q_7^3) \leq 12$.

1 Introduction

Write $[n] = \{0, 1, \dots, n-1\}$. The three-dimensional queen graph Q_n^3 has vertex set $[n]^3$; two distinct vertices are adjacent when their difference is a nonzero integral multiple of a vector in $\{-1, 0, 1\}^3 \setminus \{0\}$. Up to sign there are 13 such directions: three axis directions, six face diagonals, and four space diagonals. A set $S \subseteq [n]^3$ is dominating if every vertex belongs to $\bigcup_{q \in S} N[q]$, and $\gamma(Q_n^3)$ is the minimum size of such a set.

The two-dimensional queen domination problem and its variants have a substantial literature; see [2, 3, 4, 5]. Barr and Rao introduced the higher-dimensional problem and obtained the first general domination bound [1]. The maximum degree of Q_n^3 is at most $13(n-1)$, so every closed neighborhood has size at most $13n-12$.

The first result below refines this coarse count on the inner core. It gives exact boundary formulas and a strict separation between boundary and central interior placements. The remaining sections record projection and lifting bounds and an exact integer-programming computation for $n \leq 6$.

2 Core coverage by position type

Assume $n \geq 4$, put $m = n-2$, and let

$$C = \{1, \dots, n-2\}^3.$$

For $q \in [n]^3$, define

$$\kappa(q) = |N[q] \cap C|.$$

The following parity calculation is used for face placements.

Lemma 2.1. *Let $M \geq 1$ and*

$$f(a, c) = |a - c| + |a + c - M|, \quad 0 \leq a, c \leq M.$$

Then $f(a, c) \equiv M \pmod{2}$, and hence

$$f(a, c) \geq M \pmod{2}.$$

If M is even, equality holds only at $a = c = M/2$. If M is odd, equality holds exactly when

$$(a, c) \in \left\{ \left(\frac{M-1}{2}, \frac{M-1}{2} \right), \left(\frac{M-1}{2}, \frac{M+1}{2} \right), \left(\frac{M+1}{2}, \frac{M-1}{2} \right), \left(\frac{M+1}{2}, \frac{M+1}{2} \right) \right\}.$$

Proof. Since

$$|a - c| \equiv a + c \pmod{2}, \quad |a + c - M| \equiv a + c + M \pmod{2},$$

the congruence and lower bound follow. When M is even, equality requires $a = c$ and $a + c = M$, giving $a = c = M/2$. When M is odd, equality means $f(a, c) = 1$. This occurs either when $a = c$ and $|2a - M| = 1$, or when $a + c = M$ and $|a - c| = 1$, which gives the four listed pairs. $\square$

Theorem 2.2 (Core coverage). *Let $n \geq 4$, $m = n - 2$, and*

$$\varepsilon_m = (m - 1) \pmod{2}.$$

Then the following statements hold.

- (i) *If q is a corner, then $\kappa(q) = m$.*
- (ii) *If q lies on an edge but is not a corner, then $\kappa(q) = 2m - 1$.*
- (iii) *If q lies on a face but on no edge, then*

$$\kappa(q) \leq 5m - 4 - \varepsilon_m.$$

Equality holds at the face center when m is odd and at the four face vertices nearest the geometric center when m is even.

- (iv) *If $q \in C$, then $\kappa(q) \leq 13m - 12$. When m is odd, the central vertex attains equality. When m is even, each of the eight central vertices has coverage $13m - 18$.*

In particular,

$$\max_{q \in [n]^3 \setminus C} \kappa(q) < \max_{q \in C} \kappa(q).$$

Proof. The 13 queen lines through a vertex are pairwise disjoint away from that vertex.

For a corner, take $q = (0, 0, 0)$. Only the direction $(1, 1, 1)$ enters C , and it contains m core vertices. This proves (i).

For an edge placement, take $q = (0, 0, z)$, where $1 \leq z \leq m$. The only directions entering C are

$$(1, 1, 0), \quad (1, 1, 1), \quad (1, 1, -1).$$

Their intersections with C have sizes m , $m - z$, and $z - 1$, respectively. Thus $\kappa(q) = 2m - 1$, proving (ii).

For a face placement, take $q = (0, y, z)$, where $1 \leq y, z \leq m$, and put

$$a = y - 1, \quad b = m - y, \quad c = z - 1, \quad d = m - z, \quad M = m - 1.$$

The nine directions whose first coordinate is 1 meet C . Their core-segment lengths are

$$m, a, b, c, d, \min(a, c), \min(a, d), \min(b, c), \min(b, d).$$

Using $b = M - a$, $d = M - c$, and $\min(u, v) = (u + v - |u - v|)/2$, their sum is

$$\kappa(q) = 5m - 4 - (|a - c| + |a + c - M|).$$

Part (iii), including the equality cases, follows from Lemma 2.1.

If $q \in C$, each queen line contains at most m core vertices. Since the 13 segments share only q ,

$$\kappa(q) \leq 13(m - 1) + 1 = 13m - 12.$$

For odd m , every line through the central vertex has length m inside C . For even m , each central vertex lies on seven segments of length m and six of length $m - 1$, giving

$$7m + 6(m - 1) - 12 = 13m - 18.$$

This proves (iv).

The largest boundary value is at most $5m - 4 - \varepsilon_m$. A central interior vertex has coverage at least $13m - 18$, and

$$(13m - 18) - (5m - 4 - \varepsilon_m) = 8m - 14 + \varepsilon_m > 0$$

for $m \geq 2$. The strict separation follows. $\square$

Corollary 2.3 (Octahedral reduction). *Let O_h be the order-48 group generated by coordinate permutations and reflections $x_i \mapsto n - 1 - x_i$. Put*

$$F(n) = \left\{ (x, y, z) \in [n]^3 : 0 \leq x \leq y \leq z \leq \frac{n-1}{2} \right\}.$$

Every nonempty set $S \subseteq [n]^3$ is O_h -equivalent to a set meeting $F(n)$. In particular, every dominating-set orbit has such a representative.

Proof. Choose $q \in S$. Reflections move each coordinate of q into $[0, (n-1)/2]$, and a coordinate permutation places the three values in nondecreasing order. The resulting image of S meets $F(n)$. $\square$

For $n = 5$, the representative corner, edge, face, and interior placements

$$(0, 0, 0), \quad (0, 0, 2), \quad (0, 2, 2), \quad (2, 2, 2)$$

have core coverage 3, 5, 11, 27, respectively.

3 Bounds

Theorem 3.1 (Projection, volume, and lifting bounds). *For every $n \geq 1$,*

$$\gamma(Q_n^3) \geq \left\lceil \frac{n^3}{13n - 12} \right\rceil, \quad \gamma(Q_n^3) \geq \gamma(Q_n^2),$$

and

$$\gamma(Q_n^3) \leq n\gamma(Q_n^2) \leq \frac{69}{133}n^2 + O(n).$$

Proof. The first inequality follows because a closed neighborhood has size at most $13n - 12$.

For the projection bound, let S dominate Q_n^3 and define

$$P(S) = \{(x, y) : (x, y, z) \in S \text{ for some } z\}.$$

Projecting any three-dimensional queen move onto the first two coordinates gives either a two-dimensional queen move or the zero move. Hence $P(S)$ dominates Q_n^2 , and

$$|S| \geq |P(S)| \geq \gamma(Q_n^2).$$

For the upper bound, place a minimum dominating set of Q_n^2 in each of the n horizontal layers. This gives $\gamma(Q_n^3) \leq n\gamma(Q_n^2)$. The estimate

$$\gamma(Q_n^2) \leq \frac{69}{133}n + O(1)$$

is due to Östergård and Weakley [4]. □

Corollary 3.2.

$$\gamma(Q_n^3) = \Theta(n^2).$$

Proof. The volume lower bound is $(1/13 + o(1))n^2$, while the lifting bound is $O(n^2)$. □

4 Computational certification

4.1 Integer programming and symmetry

Let A be the adjacency matrix of Q_n^3 . Its domination number is the optimum of

$$\min \left\{ \mathbf{1}^\top x : (A + I)x \geq \mathbf{1}, \quad x \in \{0, 1\}^{n^3} \right\}. \quad (1)$$

Enumerating the maximal board segment through each vertex in each of the 13 directions constructs the adjacency relation in $O(n^4)$ time. The implementation and solver configuration are archived in [6].

The computation canonically labels candidate sets under O_h and retains one representative from each orbit. To prove that $k - 1$ queens do not suffice, the reduced feasibility problem is partitioned by the lexicographically first queen. In each branch that queen is fixed and all earlier vertices are forbidden. The branches are disjoint and exhaustive; infeasibility of every branch proves the lower bound k .

Table 1: Computed values and representative placements.

n	$\gamma(Q_n^3)$	Status	Representative placement
1	1	Exact	(0, 0, 0)
2	1	Exact	(1, 0, 0)
3	1	Exact	(1, 1, 1)
4	4	Exact	(1, 0, 3), (1, 1, 0), (1, 2, 0), (1, 3, 3)
5	6	Exact	(1, 0, 3), (1, 1, 0), (1, 3, 4), (1, 4, 1), (2, 2, 2), (3, 2, 2)
6	8	Exact	(2, 2, 2), (2, 2, 3), (2, 3, 2), (2, 3, 3), (3, 2, 2), (3, 2, 3), (3, 3, 2), (3, 3, 3)
7	10–12	Bounds	(0, 4, 3), (0, 6, 6), (1, 1, 5), (2, 3, 0), (2, 4, 0), (3, 0, 2), (3, 6, 4), (4, 3, 6), (4, 6, 4), (5, 0, 1), (6, 2, 3), (6, 5, 1)

4.2 Verification and results

Every reported placement was checked independently from the geometric definition of adjacency rather than from the constraint matrix in (1). The optimization used Gurobi [7]; for $n \leq 6$, the solver’s lower and upper bounds coincided.

For $n = 6$, the central $2 \times 2 \times 2$ block is optimal; the proof completed in approximately five seconds. For $n = 7$, a 48-hour run established the lower bound 10, while the displayed placement gives the upper bound 12.

5 Conclusion

Core coverage exhibits a strict boundary penalty: no boundary queen covers as much of the inner core as a central interior queen. The projection, volume, and lifting arguments determine the quadratic order of $\gamma(Q_n^3)$, and symmetry-reduced integer programming gives the exact values through $n = 6$.

The leading constant remains open. The present bounds leave the interval from 1/13 to 69/133; improving either endpoint requires information beyond the maximum-neighborhood count or independent layer lifting.

References

- [1] J. Barr and S. Rao, *The n -queens problem in higher dimensions*, arXiv:0712.2309 (2007). <https://doi.org/10.48550/arXiv.0712.2309>
- [2] E. J. Cockayne, Chessboard domination problems, *Discrete Mathematics* 86 (1990), 13–20.
- [3] D. Finozhenok and W. D. Weakley, An improved lower bound for domination numbers of the queen’s graph, *Australasian Journal of Combinatorics* 37 (2007), 295–300.
- [4] P. R. J. Östergård and W. D. Weakley, Values of domination numbers of the queen’s graph, *The Electronic Journal of Combinatorics* 8(1) (2001), Paper R29.
- [5] W. D. Weakley, Queen domination of even square boards, *The Electronic Journal of Combinatorics* 29(2) (2022), Paper P2.50.
- [6] *3D queen domination: computational certification code*, Zenodo (2026). <https://doi.org/10.5281/zenodo.19412672>
- [7] Gurobi Optimization, LLC, *Gurobi Optimizer Reference Manual*, 2026. <https://www.gurobi.com>